POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 3022 | Semester III | Theory | 4 Credits

Fluid Mechanics and Hydraulic Machines

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 3022 as **Fluid Mechanics and Hydraulic Machines** in Semester III for Mechanical Engineering. The official SITTTR Mechanical Engineering programme server was unavailable during this cycle. With user approval, explanatory coverage is based on a reliable public diploma curriculum for this subject and is not presented as recovered official SITTTR Course 3022 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment entries are provisional handbook placeholders aligned with adjacent Revision 2026 Semester III theory courses because official Course 3022 metadata was not retrievable. Module coverage, activities and question prompts are transparent study aids derived from the cited public diploma curriculum and must be reconciled with official SITTTR material when available.

Verified source note: The SITTTR Mechanical Engineering request repeatedly failed due TLS server termination during this cycle. The user-approved public curriculum used to structure explanations is linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

This course connects fluid properties and flow laws to pressure/flow measurement, pipe systems, pumps, turbines and hydro-pneumatic devices. It develops diagram-led reasoning, SI-unit discipline and safe selection of fluid machinery.

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Prerequisite foundation

Engineering mathematics, mechanics, basic physics and SI units.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain fluid properties, pressure and basic flow behaviour.
2. Apply continuity, energy and momentum principles to elementary flow situations.
3. Describe pipe flow, flow measurement and hydraulic losses.
4. Select and explain pumps, turbines and common hydro-pneumatic devices.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain fluid properties, pressure and fluid-static behaviour.	Understand
CO2	Apply continuity, energy, momentum and Bernoulli concepts to simple flow analysis.	Apply
CO3	Explain pipe-flow losses and select suitable flow-measuring methods.	Apply
CO4	Describe selection, working and performance of pumps, turbines and fluid-power devices.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Fluid Properties and Fluid Statics	Fluid classification; density, specific weight, viscosity and pressure; hydrostatic laws and pressure measurement.	Approx. 14 hours	CO1	Module 1
2	Fluid Flow, Energy and Measurement	Flow patterns, continuity, momentum, Bernoulli relation and devices such as Pitot tube, Venturimeter, nozzle, orifice, rotameter and notch.	Approx. 16 hours	CO2	Module 2
3	Pipe Flow and Hydraulic Losses	Reynolds number, laminar/turbulent flow, friction factor, Darcy relation, major/minor losses, water hammer and pipe selection.	Approx. 14 hours	CO3	Module 3
4	Pumps, Turbines and Fluid-Power Devices	Centrifugal/reciprocating/positive-displacement pumps; Pelton/Francis/Kaplan turbines; characteristics; hydraulic and pneumatic elements/devices.	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Fluid Properties and Fluid Statics

Fluid classification; density, specific weight, viscosity and pressure; hydrostatic laws and pressure measurement.

CO1 · Approx. 14 hours

Fluid Flow, Energy and Measurement

Flow patterns, continuity, momentum, Bernoulli relation and devices such as Pitot tube, Venturimeter, nozzle, orifice, rotameter and notch.

CO2 · Approx. 16 hours

Pipe Flow and Hydraulic Losses

Reynolds number, laminar/turbulent flow, friction factor, Darcy relation, major/minor losses, water hammer and pipe selection.

CO3 · Approx. 14 hours

Pumps, Turbines and Fluid-Power Devices

Centrifugal/reciprocating/positive-displacement pumps; Pelton/Francis/Kaplan turbines; characteristics; hydraulic and pneumatic elements/devices.

CO4 · Approx. 16 hours

MODULE 1

Fluid Properties and Fluid Statics

Fluid classification; density, specific weight, viscosity and pressure; hydrostatic laws and pressure measurement.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

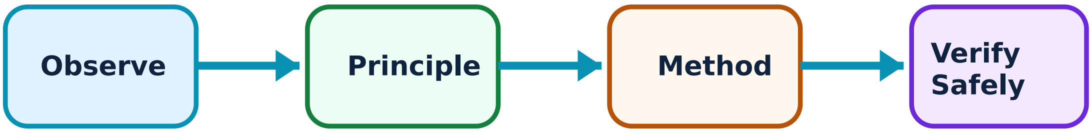
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Fluid Properties and Fluid Statics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Fluid properties and units−

Density, specific weight, specific gravity and viscosity describe fluid behaviour. SI-unit control is essential because pressure, force, head and flow rate are linked in all later calculations.

Study and application method

Write quantity, symbol and SI unit before substitution; check dimensional consistency and relate one property to a practical effect such as pressure loss or buoyancy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write quantity, symbol and SI unit before substitution; check dimensional consistency and relate one property to a practical effect such as pressure loss or buoyancy.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not use mass, weight and density as interchangeable quantities.

2. Hydrostatic pressure−

A fluid at rest transmits pressure normally to surfaces, and pressure varies with depth for a fluid of nearly constant density. Gauge and absolute pressure use different reference levels.

Study and application method

Draw a pressure reference line, identify the datum and fluid column, then calculate using consistent depth/density/gravity units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a pressure reference line, identify the datum and fluid column, then calculate using consistent depth/density/gravity units.

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Common mistake / precaution: Do not mix gauge and absolute pressure without stating the atmospheric reference.

3. Manometers and pressure gauges−

Manometers infer pressure from liquid-column difference; mechanical gauges convert pressure to a readable mechanical displacement. Instrument selection depends on range, fluid compatibility, accuracy and field use.

Study and application method

Sketch the instrument, mark interfaces/levels, traverse pressure changes systematically and state one selection criterion.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the instrument, mark interfaces/levels, traverse pressure changes systematically and state one selection criterion.

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Common mistake / precaution: A manometer reading must be interpreted with the correct fluid densities and level directions.

Mechanical-work calculator

Force (N)

10

Displacement (m)

2

Enter valid values and choose Calculate.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Fluid Flow, Energy and Measurement

Flow patterns, continuity, momentum, Bernoulli relation and devices such as Pitot tube, Venturimeter, nozzle, orifice, rotameter and notch.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

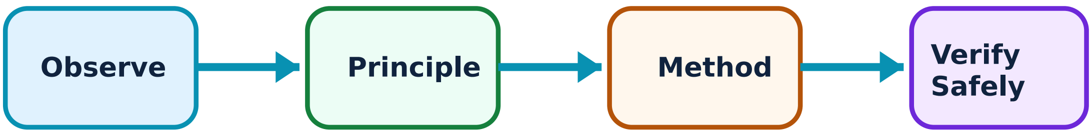
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Fluid Flow, Energy and Measurement: identify the system, state its principle, follow the correct method and verify the result safely.

1. Flow classification and continuity−

Fluid flow may be steady/unsteady, uniform/non-uniform, laminar/turbulent or compressible/incompressible. Continuity expresses mass conservation and relates area, velocity and discharge for a control volume.

Study and application method

Choose a control volume, label inlet/outlet areas and velocities, state assumptions and apply conservation of mass with units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Choose a control volume, label inlet/outlet areas and velocities, state assumptions and apply conservation of mass with units.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not apply incompressible continuity unchanged to a strongly compressible-flow situation.

2. Energy, momentum and Bernoulli principle−

Energy and momentum relations connect pressure, velocity, elevation and force effects in flowing fluid. Bernoulli’s relation is an idealised energy statement with assumptions and practical limitations.

Study and application method

Draw a streamline/control volume, list pressure/velocity/elevation at stations, state assumptions and include losses or correction only when specified.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a streamline/control volume, list pressure/velocity/elevation at stations, state assumptions and include losses or correction only when specified.

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Common mistake / precaution: Do not use Bernoulli’s equation across pumps/turbines or major loss regions without adding the relevant energy terms.

3. Flow measurement devices−

Pitot tubes, Venturimeters, nozzles, orifices, rotameters and notches measure different flow conditions. Their selection depends on fluid, range, pressure loss, installation and required accuracy.

Study and application method

State principle, construction, measured variable and application; compare two devices through a selection table before making a recommendation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State principle, construction, measured variable and application; compare two devices through a selection table before making a recommendation.

> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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> **Common mistake / precaution:** A flowmeter needs correct installation/orientation and calibration; a correct formula cannot compensate for poor setup.

> **Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Pipe Flow and Hydraulic Losses

Reynolds number, laminar/turbulent flow, friction factor, Darcy relation, major/minor losses, water hammer and pipe selection.

Approx. 14 hours | **CO3** | **Understand · Apply**

Learning goals

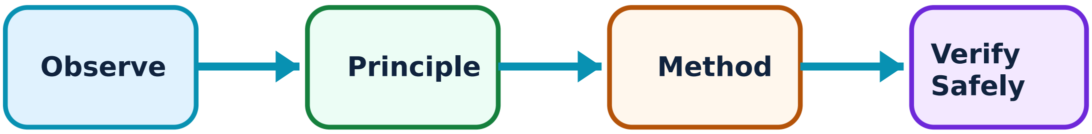
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Pipe Flow and Hydraulic Losses: identify the system, state its principle, follow the correct method and verify the result safely.

1. Reynolds number and flow regime−

Reynolds number compares inertial and viscous effects and helps characterise flow regime. It guides use of friction relations and interpretation of pipe-flow behaviour.

Study and application method

Use consistent density, velocity, diameter and viscosity units; calculate the dimensionless number and state the corresponding regime cautiously.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use consistent density, velocity, diameter and viscosity units; calculate the dimensionless number and state the corresponding regime cautiously.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Transition range and surface condition matter; do not treat one numerical boundary as universal.

2. Major and minor head losses−

Friction along pipe length creates major head loss; fittings, valves, bends and entrances create local/minor losses. Both reduce available pressure/energy in a system.

Study and application method

Draw the complete pipeline, list lengths/diameters/fittings, calculate each loss using the specified model and tabulate the total head loss.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw the complete pipeline, list lengths/diameters/fittings, calculate each loss using the specified model and tabulate the total head loss.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not neglect fittings merely because each is small; many fittings can contribute significant loss.

3. Water hammer and pipe selection−

Rapid change in flow can create pressure surges called water hammer. Pipe selection considers fluid, pressure rating, size, material, loss, cost and safety requirements.

Study and application method

Explain cause → effect → mitigation, then compare candidate pipes using a stated engineering criterion table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain cause → effect → mitigation, then compare candidate pipes using a stated engineering criterion table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Sudden valve operation can be hazardous; do not treat pressure transient as a routine steady-flow result.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Pumps, Turbines and Fluid-Power Devices

Centrifugal/reciprocating/positive-displacement pumps; Pelton/Francis/Kaplan turbines; characteristics; hydraulic and pneumatic elements/devices.

Approx. 16 hours | CO4 | Understand · Apply

Learning goals

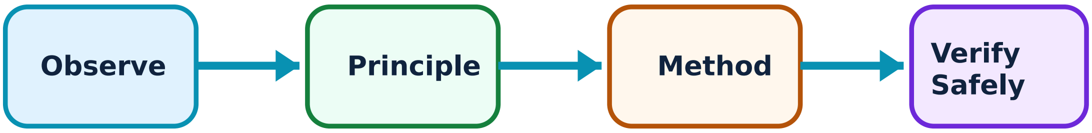
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Pumps, Turbines and Fluid-Power Devices: identify the system, state its principle, follow the correct method and verify the result safely.

1. Pumps and performance characteristics−

Pumps transfer mechanical energy to fluid. Centrifugal, reciprocating, submersible and positive-displacement pumps differ in construction, discharge behaviour and application. Head, discharge, power, efficiency and characteristic curves guide selection.

Study and application method

Identify duty point, fluid properties and required head/discharge; select pump type, explain working and interpret one performance curve qualitatively.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify duty point, fluid properties and required head/discharge; select pump type, explain working and interpret one performance curve qualitatively.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: A centrifugal pump may require priming; never start or select equipment without following manufacturer procedures.

2. Hydraulic turbines−

Turbines convert hydraulic energy to shaft output. Pelton, Francis and Kaplan turbines suit different head/discharge conditions and use impulse or reaction principles.

Study and application method

Classify by energy conversion, draw a flow path, state one suitable site condition and compare runner/flow features in a table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Classify by energy conversion, draw a flow path, state one suitable site condition and compare runner/flow features in a table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not select a turbine by output alone; available head and discharge are decisive.

3. Hydraulic and pneumatic devices−

Cylinders, valves, manifolds and devices such as presses, lifts, rams and accumulators use controlled fluid pressure for force/motion. Symbols and labelled schematics support communication and troubleshooting.

Study and application method

Read the circuit/schematic from energy source through control to actuator, identify component function and state a safe isolation/inspection step.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Read the circuit/schematic from energy source through control to actuator, identify component function and state a safe isolation/inspection step.

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Common mistake / precaution: Fluid-power systems can retain stored pressure; safe isolation is essential before maintenance.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Fluid Properties and Fluid Statics

Use the concept flow to revise observation, principle, method and verification.

Module 2: Fluid Flow, Energy and Measurement

Use the concept flow to revise observation, principle, method and verification.

Module 3: Pipe Flow and Hydraulic Losses

Use the concept flow to revise observation, principle, method and verification.

Module 4: Pumps, Turbines and Fluid-Power Devices

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a property-and-unit sheet and solve a pressure/manometer problem with a labelled datum.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare flowmeters for a chosen industrial fluid-service case using selection criteria.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a pipe-system loss table including straight pipe and fittings, then explain one water-hammer mitigation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare pump and turbine types against head/discharge/application; draw one safe hydraulic circuit with named components.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Fluid Properties and Fluid Statics

Explain Fluid properties and units and state one correct method, application or precaution.−

Density, specific weight, specific gravity and viscosity describe fluid behaviour. SI-unit control is essential because pressure, force, head and flow rate are linked in all later calculations.

Answer framework: Write quantity, symbol and SI unit before substitution; check dimensional consistency and relate one property to a practical effect such as pressure loss or buoyancy.

Important point: Do not use mass, weight and density as interchangeable quantities.

Explain Hydrostatic pressure and state one correct method, application or precaution.−

A fluid at rest transmits pressure normally to surfaces, and pressure varies with depth for a fluid of nearly constant density. Gauge and absolute pressure use different reference levels.

Answer framework: Draw a pressure reference line, identify the datum and fluid column, then calculate using consistent depth/density/gravity units.

Important point: Do not mix gauge and absolute pressure without stating the atmospheric reference.

Explain Manometers and pressure gauges and state one correct method, application or precaution.−

Manometers infer pressure from liquid-column difference; mechanical gauges convert pressure to a readable mechanical displacement. Instrument selection depends on range, fluid compatibility, accuracy and field use.

Answer framework: Sketch the instrument, mark interfaces/levels, traverse pressure changes systematically and state one selection criterion.

Important point: A manometer reading must be interpreted with the correct fluid densities and level directions.

Module 2 — Fluid Flow, Energy and Measurement

Explain Flow classification and continuity and state one correct method, application or precaution.−

Fluid flow may be steady/unsteady, uniform/non-uniform, laminar/turbulent or compressible/incompressible. Continuity expresses mass conservation and relates area, velocity and discharge for a control volume.

Answer framework: Choose a control volume, label inlet/outlet areas and velocities, state assumptions and apply conservation of mass with units.

Important point: Do not apply incompressible continuity unchanged to a strongly compressible-flow situation.

Explain Energy, momentum and Bernoulli principle and state one correct method, application or precaution.−

Energy and momentum relations connect pressure, velocity, elevation and force effects in flowing fluid. Bernoulli's relation is an idealised energy statement with assumptions and practical limitations.

Answer framework: Draw a streamline/control volume, list pressure/velocity/elevation at stations, state assumptions and include losses or correction only when specified.

Important point: Do not use Bernoulli's equation across pumps/turbines or major loss regions without adding the relevant energy terms.

Explain Flow measurement devices and state one correct method, application or precaution.−

Pitot tubes, Venturimeters, nozzles, orifices, rotameters and notches measure different flow conditions. Their selection depends on fluid, range, pressure loss, installation and required accuracy.

Answer framework: State principle, construction, measured variable and application; compare two devices through a selection table before making a recommendation.

Important point: A flowmeter needs correct installation/orientation and calibration; a correct formula cannot compensate for poor setup.

Module 3 — Pipe Flow and Hydraulic Losses

Explain Reynolds number and flow regime and state one correct method, application or precaution.−

Reynolds number compares inertial and viscous effects and helps characterise flow regime. It guides use of friction relations and interpretation of pipe-flow behaviour.

Answer framework: Use consistent density, velocity, diameter and viscosity units; calculate the dimensionless number and state the corresponding regime cautiously.

Important point: Transition range and surface condition matter; do not treat one numerical boundary as universal.

Explain Major and minor head losses and state one correct method, application or precaution.−

Friction along pipe length creates major head loss; fittings, valves, bends and entrances create local/minor losses. Both reduce available pressure/energy in a system.

Answer framework: Draw the complete pipeline, list lengths/diameters/fittings, calculate each loss using the specified model and tabulate the total head loss.

Important point: Do not neglect fittings merely because each is small; many fittings can contribute significant loss.

Explain Water hammer and pipe selection and state one correct method, application or precaution.−

Rapid change in flow can create pressure surges called water hammer. Pipe selection considers fluid, pressure rating, size, material, loss, cost and safety requirements.

Answer framework: Explain cause → effect → mitigation, then compare candidate pipes using a stated engineering criterion table.

Important point: Sudden valve operation can be hazardous; do not treat pressure transient as a routine steady-flow result.

Module 4 — Pumps, Turbines and Fluid-Power Devices

Explain Pumps and performance characteristics and state one correct method, application or precaution.−

Pumps transfer mechanical energy to fluid. Centrifugal, reciprocating, submersible and positive-displacement pumps differ in construction, discharge behaviour and application. Head, discharge, power, efficiency and characteristic curves guide selection.

Answer framework: Identify duty point, fluid properties and required head/discharge; select pump type, explain working and interpret one performance curve qualitatively.

Important point: A centrifugal pump may require priming; never start or select equipment without following manufacturer procedures.

Explain Hydraulic turbines and state one correct method, application or precaution.−

Turbines convert hydraulic energy to shaft output. Pelton, Francis and Kaplan turbines suit different head/discharge conditions and use impulse or reaction principles.

Answer framework: Classify by energy conversion, draw a flow path, state one suitable site condition and compare runner/flow features in a table.

Important point: Do not select a turbine by output alone; available head and discharge are decisive.

Explain Hydraulic and pneumatic devices and state one correct method, application or precaution.−

Cylinders, valves, manifolds and devices such as presses, lifts, rams and accumulators use controlled fluid pressure for force/motion. Symbols and labelled schematics support communication and troubleshooting.

Answer framework: Read the circuit/schematic from energy source through control to actuator, identify component function and state a safe isolation/inspection step.

Important point: Fluid-power systems can retain stored pressure; safe isolation is essential before maintenance.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Fluid Properties and Fluid Statics.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Fluid Flow, Energy and Measurement.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Pipe Flow and Hydraulic Losses.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Pumps, Turbines and Fluid-Power Devices.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Fluid classification; density, specific weight, viscosity and pressure; hydrostatic laws and pressure measurement..
2. Develop a complete answer or practical plan for: Flow patterns, continuity, momentum, Bernoulli relation and devices such as Pitot tube, Venturimeter, nozzle, orifice, rotameter and notch..
3. Develop a complete answer or practical plan for: Reynolds number, laminar/turbulent flow, friction factor, Darcy relation, major/minor losses, water hammer and pipe selection..
4. Develop a complete answer or practical plan for: Centrifugal/reciprocating/positive-displacement pumps; Pelton/Francis/Kaplan turbines; characteristics; hydraulic and pneumatic elements/devices..

LAST-MILE REVISION

Quick Revision

Module 1: Fluid Properties and Fluid Statics

Fluid classification; density, specific weight, viscosity and pressure; hydrostatic laws and pressure measurement.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Fluid Flow, Energy and Measurement

Flow patterns, continuity, momentum, Bernoulli relation and devices such as Pitot tube, Venturimeter, nozzle, orifice, rotameter and notch.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: Pipe Flow and Hydraulic Losses

Reynolds number, laminar/turbulent flow, friction factor, Darcy relation, major/minor losses, water hammer and pipe selection.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Pumps, Turbines and Fluid-Power Devices

Centrifugal/reciprocating/positive-displacement pumps; Pelton/Francis/Kaplan turbines; characteristics; hydraulic and pneumatic elements/devices.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Fluid properties and units	Density, specific weight, specific gravity and viscosity describe fluid behaviour.
Hydrostatic pressure	A fluid at rest transmits pressure normally to surfaces, and pressure varies with depth for a fluid of nearly constant density.
Manometers and pressure gauges	Manometers infer pressure from liquid-column difference; mechanical gauges convert pressure to a readable mechanical displacement.
Flow classification and continuity	Fluid flow may be steady/unsteady, uniform/non-uniform, laminar/turbulent or compressible/incompressible.
Energy, momentum and Bernoulli principle	Energy and momentum relations connect pressure, velocity, elevation and force effects in flowing fluid.
Flow measurement devices	Pitot tubes, Venturimeters, nozzles, orifices, rotameters and notches measure different flow conditions.
Reynolds number and flow regime	Reynolds number compares inertial and viscous effects and helps characterise flow regime.
Major and minor head losses	Friction along pipe length creates major head loss; fittings, valves, bends and entrances create local/minor losses.
Water hammer and pipe selection	Rapid change in flow can create pressure surges called water hammer.
Pumps and performance characteristics	Pumps transfer mechanical energy to fluid.
Hydraulic turbines	Turbines convert hydraulic energy to shaft output.
Hydraulic and pneumatic devices	Cylinders, valves, manifolds and devices such as presses, lifts, rams and accumulators use controlled fluid pressure for force/motion.

LEARNING RESOURCES

References

Recommended fluid-mechanics references

1. R. K. Bansal, Fluid Mechanics and Hydraulic Machines.
2. R. S. Khurmi, A Textbook of Hydraulics, Fluid Mechanics and Hydraulic Machines.
3. S. C. Gupta, Fluid Mechanics and Hydraulic Machines.

Approved public fluid-mechanics sources

- https://www.gtu.ac.in/Syllabus/New_Diploma/sem-3/Pdf/3331903.pdf
- https://priodeep.weebly.com/uploads/6/5/4/9/65495087/fluid_mechanics_and_hydraulic_r

The linked curriculum provides reliable public explanatory coverage while the official SITTTR Course 3022 material is unavailable. Verify any assessment or detailed-syllabus decision against restored official content.